

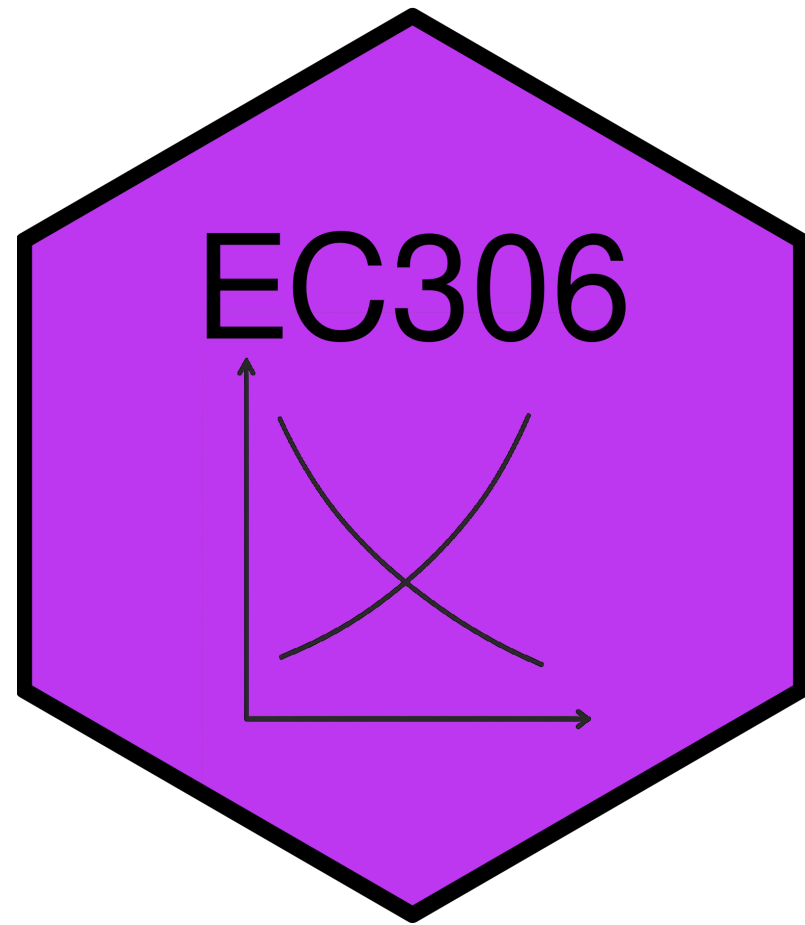
Labour Supply 3: Labour Supply Over the Life Cycle

EC306- Wages and Employment

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Fall 2025



Goals of This Section

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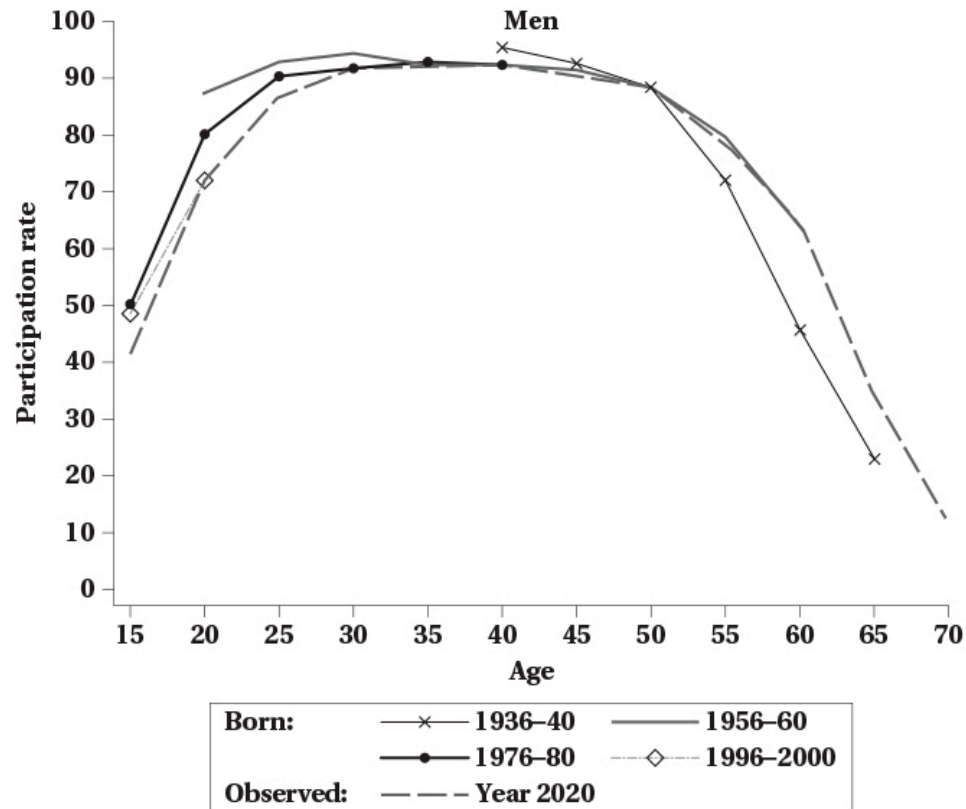
- Show patterns of labour supply over a person's lifetime
- Discuss issues comparing labour supply of people at different ages
- Introduce lifetime labour supply model
- Discuss household production
- Analyze retirement decisions

Life Cycle Labour Supply

Introduction

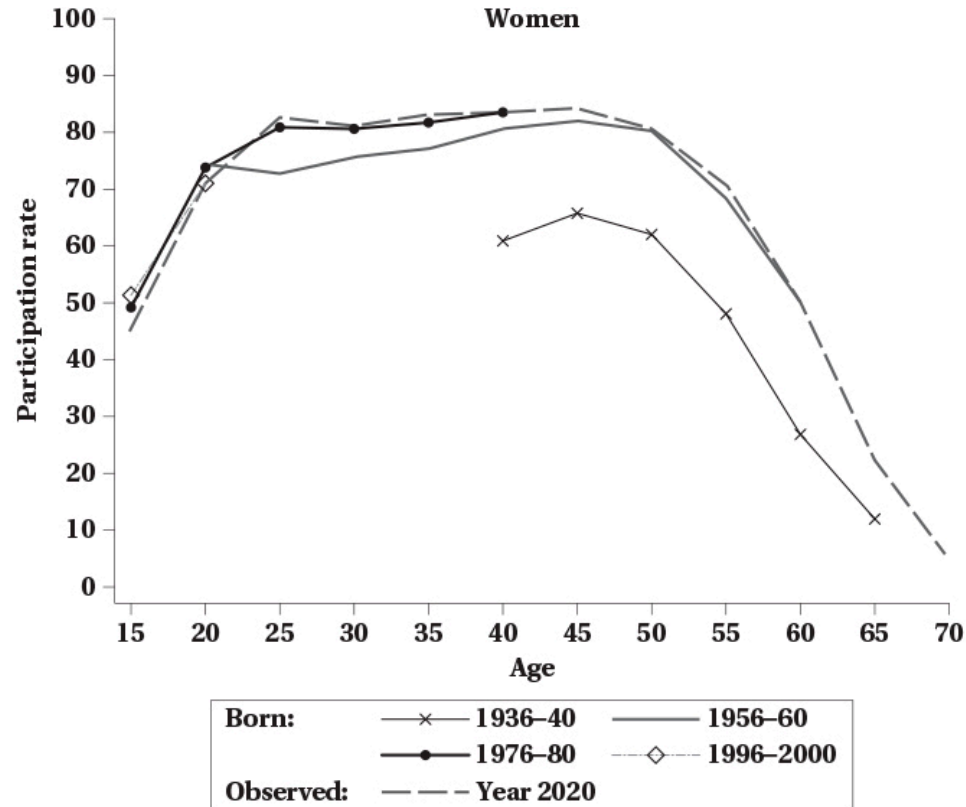
- Previous discussion focused on labour supply at a moment in time
 - Labour supply changed immediately in response to changes in wages, non-labour income, or policies
- In reality, people can plan work over their lifetime
 - Participation is highest in prime working years
 - Life events alter labour supply decisions now and in the future
 - Marriage, children, education, health, retirement
- A life cycle approach helps us understand these patterns

Life Cycle Participation for Men



- Graph follows labour supply of men born in different years
- Participation rises quickly in early 20s, peaks in late 40s, then declines
- There are differences for different birth cohorts
 - At younger ages, younger cohorts participate a bit less
 - At older ages, younger cohorts participate more

Life Cycle Participation for Women



- Women follow similar lifetime pattern
- Rates are shifted down a bit relative to men
- Participation higher for younger cohorts at all ages
 - Consistent with rising female labour force participation over time

Cohort Effects vs Age Effects

- Previous graphs highlight a technical point
- Each graph contains a line that looks at life cycle pattern in year 2020
 - Participation of people in 2020 at different ages
- Problem: people at different ages are born at different times
 - 20 year olds in 2020 born in 2000
 - 60 year olds in 2020 born in 1960
- So when we look at participation by age in a single year, we are mixing age effects and cohort effects
 - Age effects: how labour supply changes as people get older
 - Cohort effects: how labour supply differs for people born at different times

Cohort Effects vs Age Effects

- Cohort effects can be significant
 - In particular for women where it has been increasing for younger cohorts
- It is not a good approach to look at participation by age in a single year and interpret as pure age effects
- If pure age effects are of interest, better to look at participation by age for a single birth cohort
 - This is done in the graphs on the previous two slides
- Notice that the lines shift up and down for different birth cohorts
- Pattern across age is similar but there are some differences

Cohort Effects vs Age Effects

- Practically, it is difficult to do this
- Very few datasets have data on the same people over their entire life
 - Hard to track people over time
 - People die, move, refuse to respond
 - Data that does track people are often restricted access
- Another approach is to use **quasi-cohorts**
 - Group people by birth year in cross-sectional (single year) data
 - Look at how participation changes for these groups over time
 - This is what the graphs on the previous two slides do

Cohort Effects vs Age Effects

- You can do this with the Canada Census
 - Census is collected every 5 years
- Each Census collects information on age and labour force status (among many other things)
- Can follow a birth cohort across different Census years as they get older

Birth Cohort	Census Year				
	1981	1986	1991	1996	2001
1950	31	36	41	46	51
1951	30	35	40	45	50
1952	29	34	39	44	49
1953	28	33	38	43	48
1954	27	32	37	42	47
1955	26	31	36	41	46

Cohort Effects vs Age Effects

- One caveat is that in each Census year, people in a birth cohort are different people
 - Not following the exact same individuals over time
- This is generally accepted as a reasonable approximation

Cohort Effects vs Age Effects

Born in 1975

Observed in	2000	2005	2010
Age	25	30	35
Participation Rate	75%	80%	85%

Born in 1980

Observed in	2000	2005	2010
Age	20	25	30
Participation Rate	75%	80%	85%

- To illustrate problem with following ages in single Census year, examine this table
- Tracks birth cohorts across Census years
- Shows age and participation rate
- For each cohort participation rises 5 points per year
- But looking at ages in a single Census year, they appear not to change

Cohort Effects vs Age Effects

- Where do cohort effects come from? Why do we expect cohorts to be different?
- People in part are defined by their generation
 - Social attitudes change over time
 - Economic conditions and policies change over time
 - Generations experience different significant events (e.g. wars, disease)
- We therefore expect cohorts to differ in their attitudes and behaviours
 - Research has shown women born in early 1930s had higher labour supply than previous generations
 - They were influenced by their mothers working during war
 - Male children also more likely to be married to working women later in life
- Important to separate these from effects of pure aging

Dynamic Life Cycle Models

Introduction

- We now know that labour supply varies over the life cycle
- The allocation of work over time can be modeled
- Models examine the effects on labour supply of wage changes
 - Wages can change at different ages
 - They can change for a short time or permanently
 - They can be anticipated or unanticipated
- Different types of changes have different effects on labour supply

Model

- Individuals will still maximize utility over consumption and leisure subject to a budget constraint
- The utility is a function of consumption and leisure over the lifetime

$$u = U(C_1, C_2, \dots, C_N, L_1, L_2, \dots, L_N)$$

- Here $U(.)$ is a function
 - e.g. could be logarithmic, cobb-douglas, etc.
- To define the budget constraint, recognize that we can consume and work in each period
- In theory, we can consume more than our income in a period by borrowing
 - We can also save some income to consume later
- We therefore need to be conscious of the interest rate on saving and borrowing

Model

- Consider a two period model
- The budget constraint says that the present value of consumption equals the present value of income

$$C_1 + \frac{C_2}{1+r} = W_1H_1 + \frac{W_2H_2}{1+r}$$

- r is the interest rate for borrowing and saving
- This is the present (period 1) value formulation of the lifetime budget constraint
- You can also express it in future (period 2) value

$$(1+r)C_1 + C_2 = (1+r)W_1H_1 + W_2H_2$$

Model

- The ability to borrow and save means a person can consume more than their income in a period
- For example, rearranging the future value of the budget we get

$$C_2 = (1 + r)(W_1H_1 - C_1) + W_2H_2$$

- If $W_1H_1 > C_1$, then the person saved some income in period 1 to consume in period 2
 - That savings grows with interest, and they have $(1 + r)(W_1H_1 - C_1)$ to spend in period 2
- If $W_1H_1 < C_1$, then the person borrowed to consume more than their income in period 1
 - $(1 + r)(W_1H_1 - C_1)$ is negative and it reduces their income in period 2

Model

- In our model, the choice is between consumption and leisure
- To reexpress this in terms of leisure, note that $H_t = T - L_t$, so

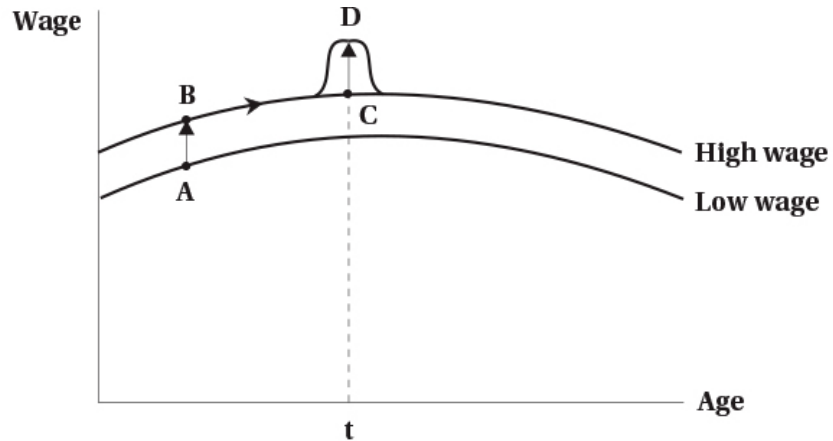
$$C_1 + \frac{C_2}{1+r} + W_1 L_1 + \frac{W_2 L_2}{1+r} = W_1 T + \frac{W_2 T}{1+r}$$

- You can consume two goods (consumption and leisure)
- The price of consumption is 1 in both periods
- The price of leisure is W_1 in period 1 and W_2 in period 2
- You consume these out of “full income”
 - Full income is the income you would have if you worked all available hours in both periods
- Period 2 values are discounted by the interest rate

Model

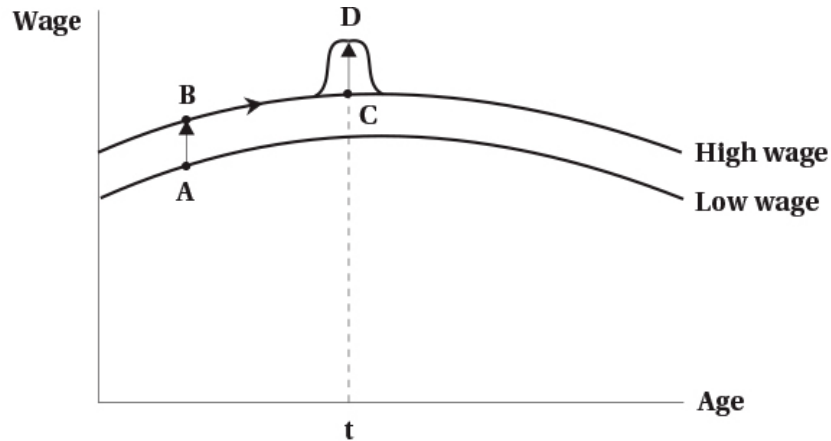
- If you maximize utility subject to the budget constraints, you get a labour supply function
 - Tells you optimal labour supply as a function of wages, interest rates, lifetime resources
- Like previous models, we can examine effects of wage changes
- There are three types
 - A permanent unanticipated change in the wage in all periods
 - An anticipated future change in the wage
 - A temporary unanticipated change in the wage in one period only
- These lead to substitution and income effects
 - More complicated because of the extra time dimension of the model
- We can see these effects in graph form

Permanent Wage Change



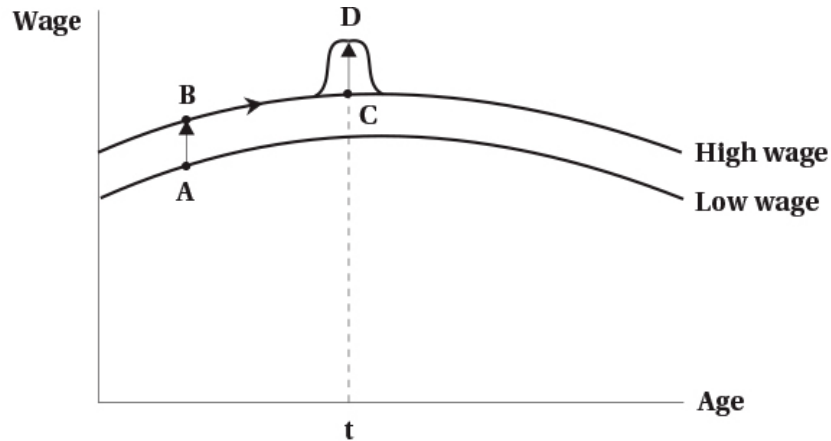
- Consider first a permanent, unanticipated change in the wage
 - Increases wage in all periods
 - Shifts wage schedule up
- Higher wage has two effects
 - Income effect: higher wage means higher lifetime income, so person consumes more leisure (less work)
 - Substitution effect: higher wage means leisure is more expensive, so person consumes less leisure (more work)
- Labour supply effect is ambiguous

Anticipated Wage Change



- An anticipated wage change moves person along their wage schedule
- There is no income effect in this case
 - Person optimized over their lifetime income
 - Lifetime income has not changed
- The substitution effect shifts leisure towards periods where wage is lower
 - If wage is higher in period 2, leisure is more expensive in period 2
 - Less leisure in period 2 and more in period 1
 - Person works more in period 2 and less in period 1

Temporary Unanticipated Wage Change



- A temporary unanticipated wage change alters wage for one period (or a small number of them) only
 - e.g. bonus pay, overtime pay, short-term contract
- Creates a temporary blip upward in wage schedule
- Two effects
 - Small Income effect: slightly higher lifetime income, so person consumes more leisure (less work)
 - Substitution effect: leisure is more expensive in the period with the higher wage, so person consumes less leisure (more work)

Comparing Effects of Wage Changes

- Strongest effect on labour supply is the anticipated wage change
 - Pure substitution effect
 - No offsetting income effect
- Next strongest is the temporary unanticipated wage change
 - Substitution effect
 - Small income effect
 - Person works more in period with higher wage
- Permanent unanticipated wage change has ambiguous effect
 - Substitution effect
 - Larger income effect
 - Ambiguous effect on labour supply

Evidence on Life Cycle Labour Supply

- Interesting experiment on bike messengers in Zurich
- Messengers paid a percentage of daily revenues
- Can choose their own total work hours
- Researchers randomly assigned messengers to
 - 25% higher wage for one month in September of one year (group A)
 - Others to 25% higher wage for one month in November of the same year (group B)
- Each acted as control group for the other
- Both get same increase in lifetime income, so difference is substitution effect
- Showed messengers shift labour supply to periods with higher wage